

Physics-Informed Digital Twin Integrating Raman Distributed Fibre Optic Sensing for Predictive Maintenance of Utility-Scale Solar Farm Inverters

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ABSTRACT

The rapid growth of utility-scale photovoltaic systems (PV) has increased the importance of reliable inverter operation, as thermal degradation and component ageing can lead to reduced energy yield and unexpected failures that scale. This research proposes a physics-informed digital twin for predictive maintenance of PV string inverters by combining a reduced-order thermal model with simulated Raman distributed temperature sensing. Spatial temperature profiles are generated under healthy and degraded operating conditions, allowing degradation-related thermal features to be identified and used for machine-learning-based fault classification. The classifier correctly identified all 14 independent simulated test cases under noiseless conditions. A subsequent noise-robustness study identified strong sensitivity to measurement noise, highlighting the need for more robust thermal feature extraction before experimental deployment. The framework aims to support condition-based maintenance, reduce unplanned downtime, and improve the reliability of utility-scale solar inverter systems that still remain a major bottleneck in the solar PV industry.

Keywords: Digital twin; Photovoltaic systems; String inverter; Raman distributed temperature sensing; Predictive maintenance; Condition monitoring; Machine learning

1. INTRODUCTION

As the world progresses towards an era marked by the transition to more sustainable energy alternatives, photovoltaic (PV) systems play a critical role to this energy transition, with solar energy now accounting for 2.8TW and having the largest growth in energy sources in 2025 [1]. However within PV plants, power electronic inverters remain among the most failure-prone components and contribute significantly to system downtime and associated energy losses that scale [2].

String inverters are widely deployed across utility-scale photovoltaic farms due to their distributed architecture, modular operation, and ability to provide independent maximum power point tracking. However, repeated thermal and electrical stresses can accelerate degradation of semiconductor devices, thermal interfaces, capacitors, solder joints, and internal heat-transfer paths [3]. These degradation mechanisms can produce measurable changes in inverter thermal behaviour before complete failure occurs.

The issue is existing condition-monitoring approaches commonly rely on discrete temperature, voltage, and current sensors. Although these methods can identify abnormal operating conditions, point-based temperature sensing provides limited spatial information and may fail to capture developing localised thermal anomalies [4,5].

2. RESEARCH AIM

This research therefore investigates a physics-informed digital twin that combines a reduced-order inverter thermal model with Raman distributed temperature sensing. By comparing predicted thermal behaviour with spatial temperature measurements, degradation-sensitive thermal residuals and features can be extracted and used to support condition monitoring and machine-learning-based degradation classification.

3. RESEARCH OBJECTIVES

The principal objectives are:

1. Develop and validate a reduced-order electrothermal digital twin of a utility-scale PV string inverter incorporating representative thermal degradation mechanisms.
2. Develop a Raman distributed temperature sensing model capable

of reconstructing spatial temperature profiles and quantifying thermal anomalies through comparison with digital-twin predictions.

3. Develop and evaluate physics-informed feature based machine-learning methods for detecting and classifying inverter thermal degradation across varying operating conditions, supporting predictive maintenance.

4. PROPOSED METHODOLOGY

The proposed framework consists of three primary components: a physics-based thermal model, a distributed photonics sensing model, and an XGBoost driven degradation-classification system.

A Cauer thermal network will represent heat generation and heat transfer between semiconductor devices, heat sinks, thermal interfaces, and the surrounding environment [6]. Model parameters will describe the dominant thermal resistances and capacitances governing inverter temperature dynamics.

Raman DTS will then be represented as a distributed measurement layer, providing temperature observations along sensing fibre positioned at thermally significant inverter locations [4,8].

Differences between digital-twin predictions and distributed sensor measurements will form residual features used to identify abnormal thermal behaviour.

5. DIGITAL TWIN SYSTEM ARCHITECTURE

The developed digital twin couples an electrothermal inverter model with Raman distributed temperature sensing (DTS). Semiconductor losses are propagated through a reduced-order Cauer thermal network to estimate the inverter temperature, which is mapped onto the sensing fibre to obtain the spatial temperature profile $T_{DTS}(x, t)$.

Thermal degradation is represented by controlled increases in the internal thermal resistance of the Cauer network [3,7]. The simulated degradation states are summarised in Table 1.

Table 1: Simulated thermal degradation states.

State	α_{deg}	ΔR_{th}
Healthy	1.00	0%
Mild	1.25	+25%
Moderate	1.50	+50%
Severe	2.00	+100%

These levels are controlled simulation perturbations rather than

universal degradation thresholds. The difference between inverter surface temperature and its distributed DTS measurement is quantified as $r_{DTS}(x, t) = T_{DTS}(x, t) - T_{surface}(t)$. Degradation-sensitive changes are then evaluated relative to the corresponding healthy baseline, allowing changes in both the magnitude and spatial distribution of the thermal response to be identified. Multiple physics-informed features are extracted from these responses for degradation classification.

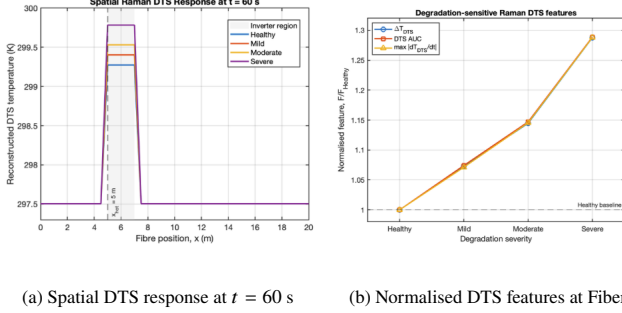


Figure 1: Raman DTS response to increasing thermal degradation: (a) spatial temperature profiles and (b) normalised thermal features relative to the healthy baseline.

6. RAMAN DISTRIBUTED TEMPERATURE SENSING

When an optical pump pulse propagates through the fibre, a small fraction of the light undergoes Raman scattering, producing lower-frequency Stokes and higher-frequency anti-Stokes backscatter. The anti-Stokes intensity is strongly temperature dependent because it depends on the thermally populated vibrational states of the fibre, while the Stokes component is comparatively less temperature sensitive. The return time identifies the position along the fibre, enabling distributed temperature measurement [4,8,11].

Temperature is reconstructed from the anti-Stokes-to-Stokes ratio,

$$R(x) = \frac{I_{AS}(x)}{I_S(x)} = C(x) \exp\left(-\frac{\Theta_R}{T(x)}\right), \quad \Theta_R = \frac{hc\Delta\tilde{\nu}}{k_B},$$

where $C(x)$ accounts for calibration and differential optical losses and $\Delta\tilde{\nu} = 440 \text{ cm}^{-1}$ is the Raman shift [8,11]. The local fibre temperature is therefore reconstructed as

$$T_{DTS}(x) = -\frac{\Theta_R}{\ln[R(x)/C(x)]}.$$

For the present degradation-classification study, a reduced-order Raman DTS model is used to generate spatial temperature measurements for the machine-learning dataset. The predicted inverter surface temperature is mapped onto a 20 m sensing fibre and passed through a fibre thermal-response model. The fibre is discretised at 0.5 m intervals, producing 41 spatial temperature channels; sub-metre Raman DTS spatial resolution of 0.5 m has been experimentally demonstrated [12].

The reduced model retains the temperature-dependent Raman response, spatial mapping, fibre thermal response, and optical attenuation while abstracting the detailed receiver electronics, enabling efficient generation of the dataset used for degradation classification.

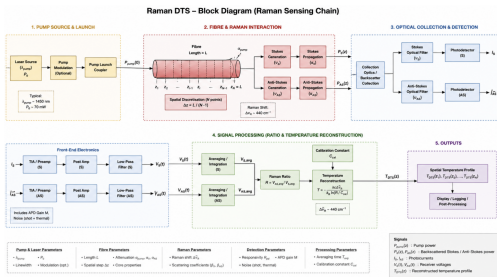


Figure 2: Higher-fidelity Raman DTS sensing chain incorporating Raman backscatter, optical detection, receiver processing, and temperature reconstruction.

A higher-fidelity Raman DTS model, shown in Fig. 2, has

also been developed to represent the complete optical and receiver chain. Future work will compare this model with the reduced-order model to quantify the effects of attenuation, receiver noise, APD gain, averaging, and calibration on temperature reconstruction and degradation-sensitive features [8,10].

7. XGBOOST ALGORITHM

Multiple physics-informed features were extracted from the Raman DTS temperature profiles to characterise the temporal and spatial thermal response of the inverter. These features were used as inputs to an XGBoost classifier to distinguish between the considered degradation states [9].

It combines multiple decision trees to produce a final prediction,

$$\hat{y}_i = \sum_{k=1}^K f_k(\mathbf{x}_i), \quad (1)$$

where $\mathbf{x}_i$ represents the extracted thermal feature vector, K is the number of trees, and $f_k(\mathbf{x}_i)$ is the prediction from the k th tree [9]. Feature-importance analysis identified temperature rise and spatial temperature gradient as the most influential features for distinguishing between the simulated degradation states.

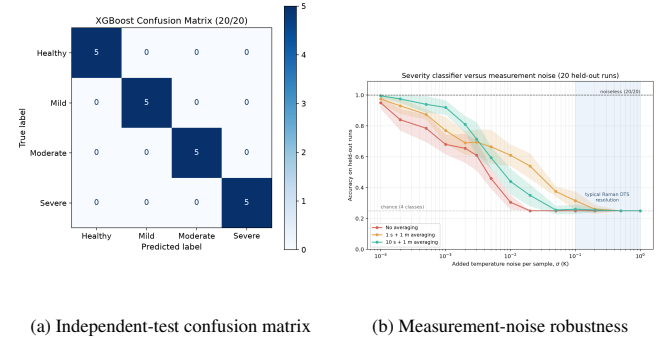


Figure 3: XGBoost degradation-classification performance: (a) confusion matrix for the independent simulated test cases and (b) classifier accuracy under increasing simulated temperature measurement noise.

8. RESULTS AND DISCUSSION

As shown in Fig. 1, increasing thermal degradation produces greater temperature rise and spatial temperature gradient within the inverter region, forming a measurable degradation signature [3,7]. These spatial thermal changes provide physics-informed features linking the simulated degradation mechanism to automated condition assessment. The XGBoost classifier correctly classified all 14 independent off-grid Simulink test cases, demonstrating clear separation between the four degradation states under noiseless simulated conditions.

Caveat is the noise study in Fig. 3, identified a key limitation. Accuracy remained above 95% only up to approximately 10^{-4} K added noise without averaging and 2×10^{-4} K with 10 s temporal and 1 m spatial averaging, before progressively approaching the four-class chance level of 25%. Although averaging improved robustness, the improvement was insufficient at higher noise levels.

The results therefore demonstrate strong degradation discrimination under ideal simulated conditions while identifying noise-robust feature extraction as a key requirement before experimental deployment.

9. NEXT STEPS

Future work will replace the noise-sensitive maximum temperature derivative with more robust heating-rate estimation and retrain the classifier under Raman-DTS-representative noise. The DTS model will also be extended to longer sensing distances and compared with the higher-fidelity optical model to quantify attenuation, receiver, and calibration effects [10]. Experimental validation will then assess generalisation and support future degradation tracking and remaining-useful-life estimation.

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